

STAT3007/7007 Deep Learning — Course Project

2026 Semester 2

Topic

a topic of your choice on deep learning

Group size

4 to 5, submit your group member names at <https://tinyurl.com/23o4njw2> by 24 Aug.

Deliverables

proposal	5%	due at 5pm on 14 Sep (submit a PDF file) submit contribution statement at https://tinyurl.com/2dnv5o92
seminar	10%	due in week 13 (submit slides and give a talk)
report	20%	due 5pm 9 Nov (submit a PDF report, and code if applicable)
reflective essay	5%	due 5pm on 10 Nov (submit a PDF file) submit contribution statement at https://tinyurl.com/2cnwkj7q

You will be assigned a group if you do not have a group by the deadline to submit your group member names. Contribution statements and reflective essay are individual submissions. Proposal, seminar slides and report are group submissions — only one group member needs to submit. Marks may be moderated based on the contribution statements.

While the use of AI tools is allowed, you should clearly state how you use the tool in your assessment deliverables.

Topics

You may choose to work on an application project, an algorithm/method project, a theory project, or any type of project that you are keen to work on, as long as the scope is substantial enough to demonstrate strong understanding of deep learning.

Application. An application project mainly involves developing a deep learning solution to a practical problem. For example, you may apply techniques like RNNs, ResNets, or GANs to a challenging practical problem.

Your analysis may involve answering some or all of the following questions below.

- (Exploratory analysis) Are the covariates correlated? How are the covariates distributed?
- (Model diagnostics) Is/Are the fitted model/models appropriate for the dataset (e.g. whether there are evidence that the model assumptions may be inappropriate)? Can we use them to draw conclusions?
- (Diagnostic analysis) What are the important factors/covariates?
- (Predictive analysis) What does the model predict on new data/situation? Does the model generalize well on new data? Can we interpret the model's prediction?
- (Prescriptive analysis) What actions should be taken based on the model?

In general, you do not need to answer all the questions above, and you may choose to investigate questions not listed above. The important thing is to make sure that you can justify the significance of the questions that you work on.

Algorithm/method. An algorithm project requires developing a new algorithm for a deep learning task. It may be about modifying some existing models/algorithms and see whether that works, but it can also be about proposing a highly novel approach. As another example, you can also perform extensive experiments to study how different methods/models compare with each other on a range of datasets (e.g. comparison of different GAN models for image generation).

Theory. A theory project focuses on theoretical aspects of deep learning. It may contain a rigorous account of a substantial theoretical topic (e.g. generalization bounds for learning algorithms), or may present new proofs or theoretical results. Such a project can be purely theoretical, but may involve some experiments on simulated or real data for validating the theory.

Data sources. Here are some sources of publicly available data that you may find helpful for your project. If you already have a problem that you want to solve by doing some statistical analysis, you can go to these data sources and search for the data you need. If you do not have a specific idea yet, it will be helpful to browse through some of the datasets to see whether you find something interesting.

- Kaggle datasets: <https://www.kaggle.com/datasets>.
- Hugging Face datasets: <https://huggingface.co/datasets>. See <https://huggingface.co/docs/datasets/> for the documentation.
- Datasets used in research papers: <https://paperswithcode.com/datasets>. This website only provides links to the homepages of various datasets. It is recommended that you search the Internet to see if there is any existing code for loading the datasets before you write your own code for doing this. For example, CIFAR-10 is top in the dataset list, and you can use the `torchvision.datasets.CIFAR10` function in the `torchvision` library to load it.
- OpenML datasets: <https://openml.org/>
- Some government datasets: <https://www.data.gov/>, <https://data.gov.au/>, <https://data.gov.sg/>.
- UCI machine learning datasets: <https://archive.ics.uci.edu/ml/datasets.php>.
- FiveThirtyEight (interactive sports and news site); <https://data.fivethirtyeight.com/>.
- BuzzFeedNews (America Internet media, news and entertainment company): <https://github.com/BuzzFeedNews>.
- Socrata (cleaned open source data sources ranging from government, business, and education data sets): <https://opendata.socrata.com/>.

Deliverables

Proposal. You should use the template provided to write your proposal. The proposal should cover the following aspects within 2 pages:

- Title and topic: write down the title of your project and explain the proposed problem.
- Significance: explain why the problem is important.
- Feasibility: justify that the project is feasible by explaining what has been done, what the planned work will be and how it will be carried out. If data is needed for your project, make sure that you have found some, as sometimes data does not come as easily as we expect.

The proposal will be assessed based on its clarity and conciseness, significance, feasibility, and novelty.

Seminar. Each group will give a talk to showcase their project. You will need to upload the slides for the talk, which should be either as .ppt file or a .pdf file. The length of each talk will be announced closer to the seminar date, but most likely will be around 10 minutes for each talk.

The talk should clearly convey to the audience the following aspects of the projects: the problem, significance of the problem, steps taken to solve the problem, main findings, limitations of the study, and conclusion.

The talk will be assessed based on whether the above points are clearly explained, and whether the claims are compellingly supported with adequate amount of technical details (See Tips section for some examples). While it is understood that a short talk cannot provide all the technical details needed for reproducibility, a lack of technical details will make your talk look like a show of opinions, rather than a scientific presentation.

Report. This is the most significant deliverable of the project.

You should use the template provided to write your report. There is some flexibility in the organization of the report, but the main text of your report is expected to cover the following aspects of your project within 12 pages: the problem, significance of the problem, steps taken to solve the problem, main findings, limitations of the study, and conclusion.

Your report will be assessed based on the following criteria.

- Clarity and conciseness: no spelling, grammar, and punctuation mistakes; content is well-organized; notations and technical statements are clear; writing is straight to the point.
- Correctness: the arguments and technical details should be sound and correct.
- Reproducibility: the report should include sufficient details so that a reader can follow the report to reproduce the presented results.
- Significance: the project can be significant in various ways, including the choice of an important societal or scientific problem, the amount of effort or creativity required to

develop the reported solution, the thoroughness of the study, the potential impact of the findings. You should highlight in your report what you believe to be significant.

- **Novelty:** demonstration of creativity by proposing new ways of formulating an application as a machine learning problem, finding out limitations of algorithms that have not been observed for the application, developing new methods to interpret the model for the application... You should highlight in your report what you believe to be novel.

For a project involving implementation, you should submit your code separately as a zip file on Blackboard.

Reflective essay. Submit a 1 page PDF file outlining your personal reflection of your most significant learning from the project. Provide an account on the major issues that you encountered, the choices that you made, what worked, what went wrong, and what could be improved. You can also discuss any practical issues encountered in the process. You should use the template provided.

The reflective essay will be assessed based on clarity and insight *supported by your personal experience*. It should not be a summary of your project report.

Tips for your presentation and report

A good project need to demonstrate good understanding of the problem, your skills in crafting a solution, rigorous and thorough evaluation of your solution, and effective communication of the significant process and findings via a presentation and a report.

The following are some specific tips that you may find helpful for achieving the above objective, particularly if you are working on an application or algorithm project. Note that it is important not to include excessive amount of details. For example, it is not interesting to fill your presentation/report with numerous training curves if you train a large number of models. Focus on getting the main messages across, and provide sufficient details to support them.

- **Problem and significance**
 - State the problem statement by setting the objectives to be achieved, with an explanation of relevant background or context needed for understanding it.
 - The problem’s significance can be demonstrated by explaining how it is related to other problems, its applications, with citations to related references.
- **Solution**
 - Provide model and training details. It need to be detailed enough so that your approach can be reproduced and correctness can be verified. A technical project shouldn’t be assessed based on whether I trust you.

For example, instead of just saying “we train a two-hidden layer convolutional network using SGD”, also provide details such as hyperparameters of the convolutional layers, activation function, loss function, learning rate, momentum,...
 - Highlight important tricks that you have used, e.g. data augmentation, design of domain-specific loss, balancing dataset, ...

- Findings

- Explain the setup for evaluating your system clearly. In particular, describe the datasets and its characteristics, and the performance measure.
- Plots of learning curves (train/test performance against iteration/epochs) are very useful for identifying obvious convergence issues.
- An overall performance of your system should be reported on a test set.

Many problems have well-defined and commonly used performance measures, e.g., we use measures like accuracy/precision/recall/F-measures for classification problems, and MSE for regression problems. Your system's test set performance should be included to reflect the solution's overall performance (including training set performance helps as well).

In cases where there is no commonly used numerical performance measure, evaluation on some well-designed cases will be indicative of the system's performance.

- A more in-depth analysis on the system's performance is to analyze when the system does well and does poorly.

For example, for classification, confusion matrix allows us to which classes are easy and which are hard, and common errors. Examining examples when the system works and fails can also provide insights on the system's strength and weakness.

- Limitations

- If a satisfactory solution has not been found, this will be the place to discuss practical difficulties, and what you would have explored for making things work if time permits.
- Clarifying limitations in your solution is very helpful for readers to better interpret your results.
- Any other limitation of the study can be discussed here as well. The main purpose is to help readers to better understand your approach and results. It also shows that you have holistic understanding of your study.

- Conclusion

- This can include a summary of significant findings, e.g. what you found to be most helpful for getting good performance, what makes the problem difficult if a good solution has not been found. This should help readers to get a good idea of what this project has achieved quickly.
- This can also include suggestions based on your findings.

Tips on project management

- Define clear tasks and responsibilities for each group member. In general, it is best to have someone put themselves forward to play the role of a project manager by taking up the responsibilities for organizing meetings, allocating tasks, leading/facilitating discussions,...

- Take small steps on a regular basis, and meet regularly to discuss progress and issues. Don't leave everything to the last minute.
- Set up communication protocols. Make good use of emails, texting apps, and communication platforms like Slack/Zulip/Discord.
- Assess whether an approach is computationally feasible before committing to it. Run small-scale experiments (e.g., use a small subset, use a small number of mini-batches) to estimate the amount of computational resources required.
- Standardize experimental setup when comparing different algorithms, and make sure everyone is following the same setup. Avoid having two people running two different algorithms on two different train-test splits when comparing the algorithms' performances.
- Someone should check the entire presentation and report for coherence and make necessary edits.